

Jingxuan Liu

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science

Expected May 2028

Bachelor of Science in Computer Science, GPA: 3.53

Expected May 2027

- Relevant Coursework: Data Structures and Algorithms, Design and Analysis of Algorithms, Machine Learning, Intro to Databases, Computer Organization and Programming, Systems and Networks, Objects and Design

TECHNICAL SKILLS

Languages: Java, Python, SQL, C/C++, Assembly, JavaScript, HTML/CSS

Frameworks: FastAPI, Django, TensorFlow/Keras, REST APIs

Databases: MySQL, MongoDB, InfluxDB

Developer Tools: Git, VS Code, Jupyter Notebook

Libraries: OpenCV, dlib, NumPy, pandas, scikit-learn, Matplotlib

EXPERIENCE

Research Intern

May 2026 – August 2026

WISH Center, Georgia Institute of Technology

Atlanta, GA

- Developed a MobileNetV3Small computer vision pipeline to classify driver eye states from video frames.
- Implemented OpenCV and dlib facial landmark preprocessing to detect faces, crop eye regions, and prepare model inputs.
- Computed PERCLOS-based fatigue metrics and generated annotated video outputs to visualize drowsiness trends over time.

Handshake AI Fellowship

August 2025 – January 2026

Handshake

Remote

- Evaluated AI model outputs for instruction following, reasoning quality, factual accuracy, and response clarity.
- Identified failure patterns in model responses and provided structured feedback with corrected examples.
- Contributed high-quality human judgments used to improve model alignment, reliability, and output quality.

PROJECTS

Driver Fatigue Detection System | Python, OpenCV, dlib, TensorFlow/Keras

May 2026 – Present

- Developed a computer vision pipeline to detect driver eye states from video frames using OpenCV, dlib facial landmarks, and TensorFlow/Keras.
- Preprocessed video frames by detecting faces, cropping eye regions, and preparing image inputs for model inference.
- Computed PERCLOS-based fatigue metrics and generated annotated video outputs to visualize drowsiness trends over time.

YouTube Engagement Prediction | Python, Machine Learning, NLP

August 2025 – December 2025

- Built machine learning models to predict YouTube engagement using comment text and video metadata.
- Engineered NLP and metadata features to capture patterns associated with popularity and audience interaction.
- Evaluated predictive models to identify features most strongly associated with video engagement.

Time Map | Django, Python, MySQL

March 2025 – May 2025

- Collaborated with a five-person team to build a full-stack task and time management web application.
- Implemented user authentication, password management, and MySQL-backed data models in Django.
- Wrote unit and integration tests to validate core workflows and improve application reliability.

Movie Store | Django, Python, MySQL

January 2025 – March 2025

- Developed a Django-based movie purchasing application with catalog browsing and simulated checkout workflows.
- Designed MySQL schemas for movie catalogs, user records, and transaction data.
- Validated purchasing logic with 100+ mock transactions to test core database and checkout behavior.

LEADERSHIP & COMMUNITY INVOLVEMENT

Supercomputing @ GT | Member

August 2024 – Present

- Participate in technical meetings on high-performance computing, parallel programming, and systems concepts.
- Apply Java and algorithmic problem-solving skills in introductory HPC coding workshops.

Artificial Intelligence @ GT | Member

August 2024 – Present

- Collaborate on AI projects involving data preprocessing, feature engineering, and machine learning models.
- Implemented Python-based ML models for team projects focused on analyzing dataset patterns.